

A Two-axis regime taxonomy

Training-time cohort
visibility

P0 / R0-T
PRIMARY
query nodes
visible during
training
cohort-visible
inference

P1 / R1-C
PRIMARY
train-only graph
cohort-visible
inference

P2 / P4 / P5
secondary /
sensitivity
regimes

P3 / R2-Q
secondary campaign
strict
single-query
inference

Inference-time query context →

B Visibility is not supervision

P0 — training graph

P1 — training graph

Fit
subjects
**labels
used**

Fit subjects
labels used

Query
subjects
unlabeled

Query
subjects
**absent during
training**

Same supervised fit set + same validation-based
selection rules.

**Intervention: graph visibility, not
diagnostic-label supervision.**

C Primary estimand and replication scope

$$\Delta\Delta\text{AUC} = (\text{AUC}_{\text{VGCN}, P0} - \text{AUC}_{\text{VGCN}, P1}) - (\text{AUC}_{\text{Parisot}, P0} - \text{AUC}_{\text{Parisot}, P1})$$

Split 666
frozen prior
audit

Split 1024
M4.5-S1
extension

Split 2024
M4.5-S1
extension

Three preselected splits = replication units
**No pooling · no cross-split mean · no
arbitrary-split invariance claim**

M4.5-S1 extension run: 1024 + 2024 only (120 cells).